

Midwest Advanced Materials are entering the magnet manufacturing and other downstream sectors. A production-linked incentive scheme worth ₹35-₹50 billion is also being prepared to boost local capacity.

Still, amid all this activity, a fundamental question remains: Is developing more rare-earth capacity the only option? India has been looking for alternatives for a long time.

As early as 2023, Tata Elxsi had announced an EV motor that used ferrite in place of rare-earth metals. In an interview with *Electronics For You*, Anush G. Nair, Senior Technology Manager at the company, detailed how the motor was designed specifically for small commercial vehicles.

"The engine has undergone testing on both a dynamo and a Tata ACE vehicle. We are currently in the conceptual testing phase. We aim to qualify it as a concept, and the design has successfully undergone various tests, all showing the motor's performance to exceed expectations," he said at that time.

As of mid-2025, the ferrite-based EV motor remains in the concept/testing stage. No public announcement indicates that it has moved into mass production.

Last month, Ola Electric began exploring ferrite-based motors for its future models and has already announced the production of magnet-free motors on its official YouTube channel. According to a report by *Business Standard*, it plans to launch these in the third quarter of the year.

Across India, both startups and legacy players are also developing rare-earth-free motor solutions. Bengaluru's Chara Technologies is working on synchronous reluctance motors (SynRM), while Conifer (Pune and Bay Area) and Chennai's Viridian Ingini Propulsion are using ferrite magnets in axial-flux and hybrid designs. Ather Energy and Sona Comstar are investigating 'light rare-earth' materials, such as cerium and samarium, while Mahindra and Uno Minda are focusing on local magnet manufacturing.

Academic R&D is also ramping up. IIT Bhubaneswar and Numeros Motors are co-developing indigenous magnet-free topologies, VNIT has unveiled a rare-earth-free motor with up to 25% cost savings, and IIT Delhi is studying designs with minimal rare-earth input. KPIT Technologies is pushing magnet-free induction motors for heavy-duty EVs.

While domestic efforts to develop rare-earth-free EV motors are underway, British company Advanced Electric Machines (AEM) has taken significant strides in commercialising a new class of electric motors that eliminate both rare-earth magnets and copper



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